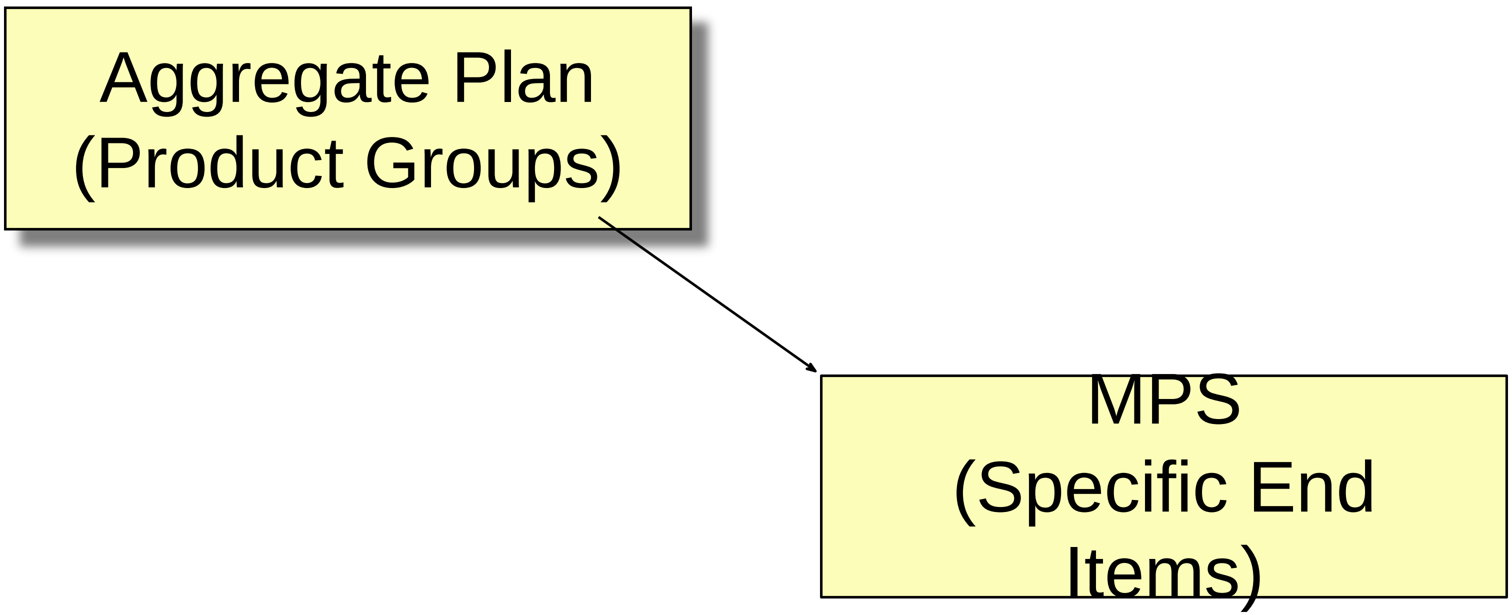
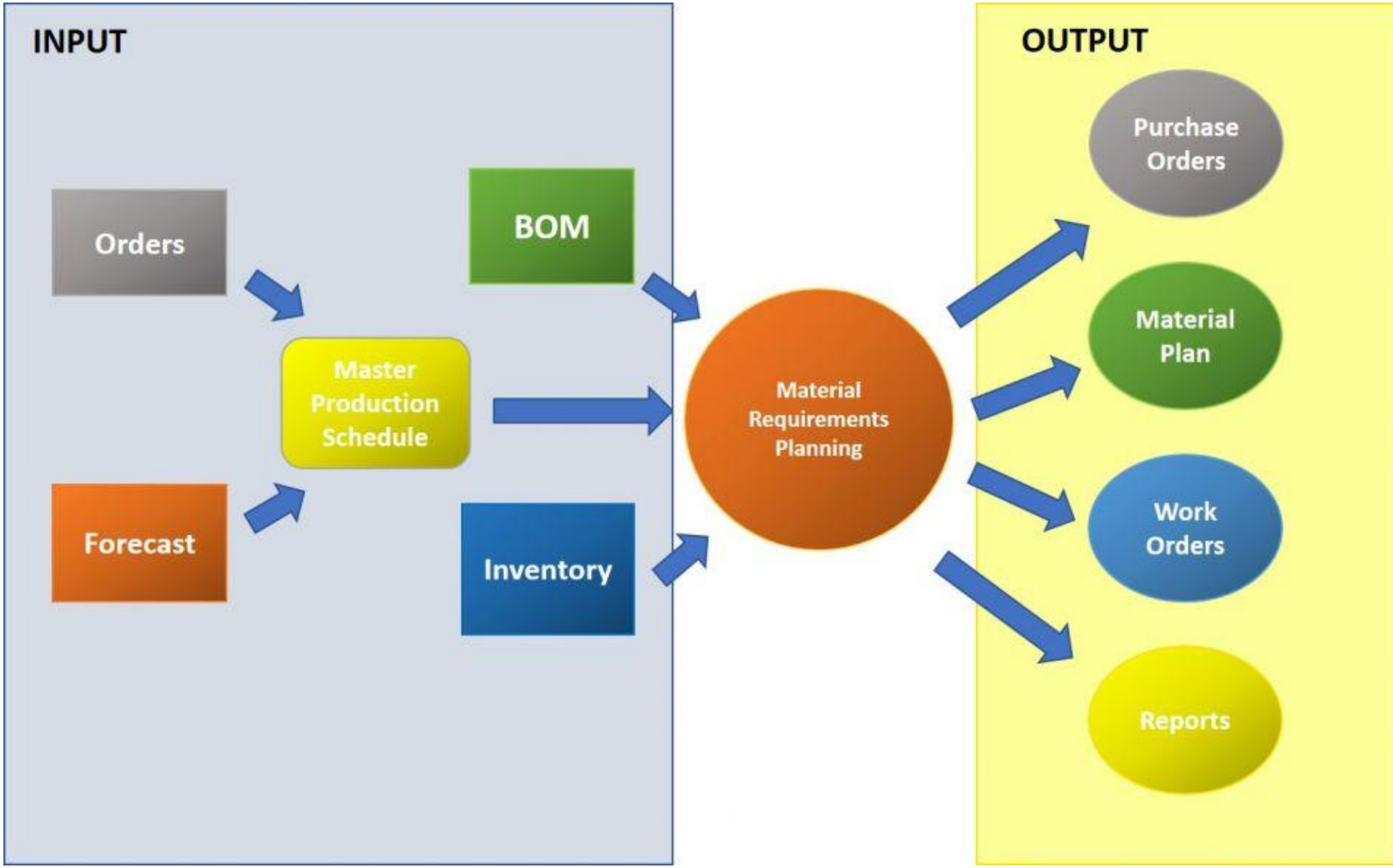
Materials Requirements Planning: Smart Production Planning

Right quantity | Right time | Right component

Master Production Schedule (MPS)

- **Time-phased plan specifying *how many* and *when* the firm plans to build each *end item***




INPUT
Orders
BOM
Master Production Schedule
Forecast
Inventory
Material Requirements Planning
OUTPUT
Purchase Orders
Material Plan
Work Orders
Reports

Material Requirements Planning

- **Materials requirements planning (MRP) is a means for determining the number of parts, components, and materials needed to produce a product (What, How much, When)**
- **MRP provides time scheduling information specifying when each of the materials, parts, and components should be ordered or produced**
- **Dependent demand drives MRP**

Material Requirements Planning System

- **Based on a master production schedule, a material requirements planning system:**
  - **Creates schedules identifying the specific parts and materials required to produce end items**
  - **Determines exact unit numbers needed**
  - **Determines the dates when orders for those materials should be released, based on lead times**

You manage a factory.

Demand: 50 units in 10 days.

When will you order components?

Problem

Early → High cost

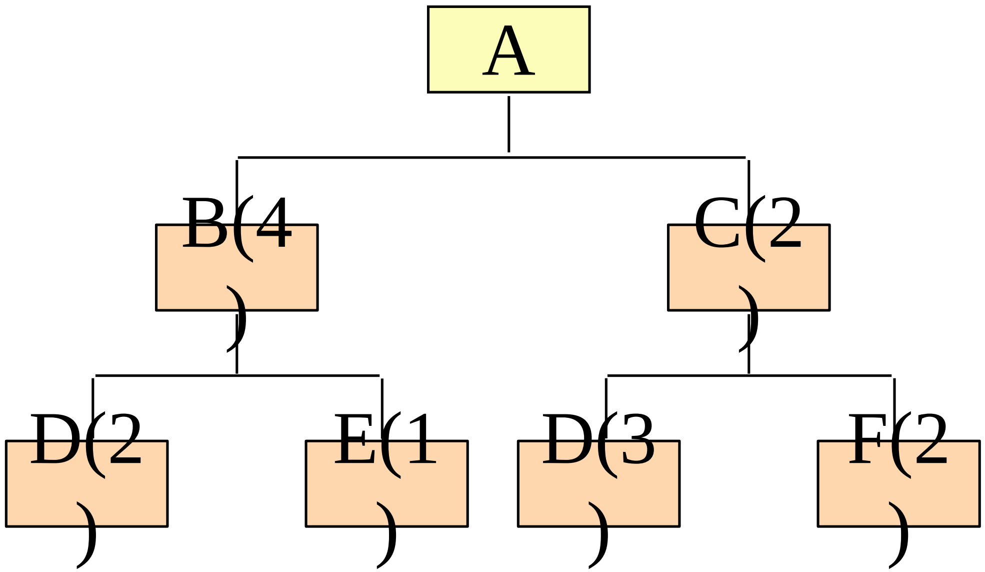
Late → Delay

Need smarter planning

Example of MRP Logic and Product Structure Tree

Given the *product structure tree* for "A" and the lead time and demand information below, provide a materials requirements plan that defines the number of units of each component and when they will be needed

Product Structure Tree for Assembly A



<u>Lead Times</u>

A.1 day
B.2 days
C.1 day
D.3 days
E.4 days
F.1 day

<u>Total Unit Demand</u>

Day 10	50 A
Day 8	20 B (Spares)
Day 6	15 D (Spares)

First, the number of units of "A" are scheduled backwards to allow for their lead time. So, in the materials requirement plan below, we have to place an order for 50 units of "A" on the 9th day to receive them on day 10.

	Day:	1	2	3	4	5	6	7	8	9	10
A	Required										50
	Order Placement									50	

LT = 1 day

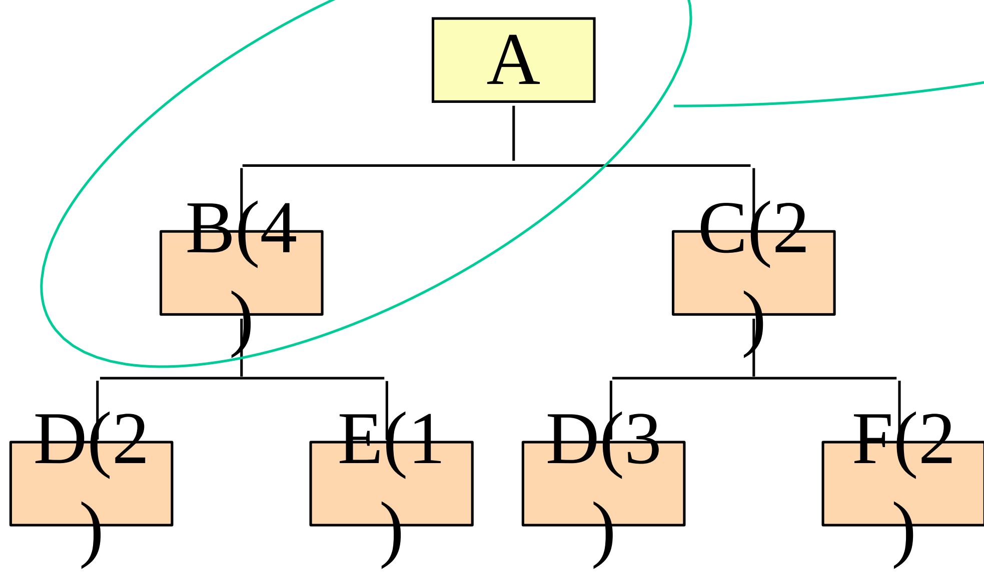
Next, we need to start scheduling the components that make up "A". In the case of component "B" we need 4 B's for each A. Since we need 50 A's, that means 200 B's. And again, we back the schedule up for the necessary 2 days of lead time.

	Day:	1	2	3	4	5	6	7	8	9	10
A	Required										50
	Order Placement									50	
B	Required								20	200	
	Order Placement						20	200			

LT = 2

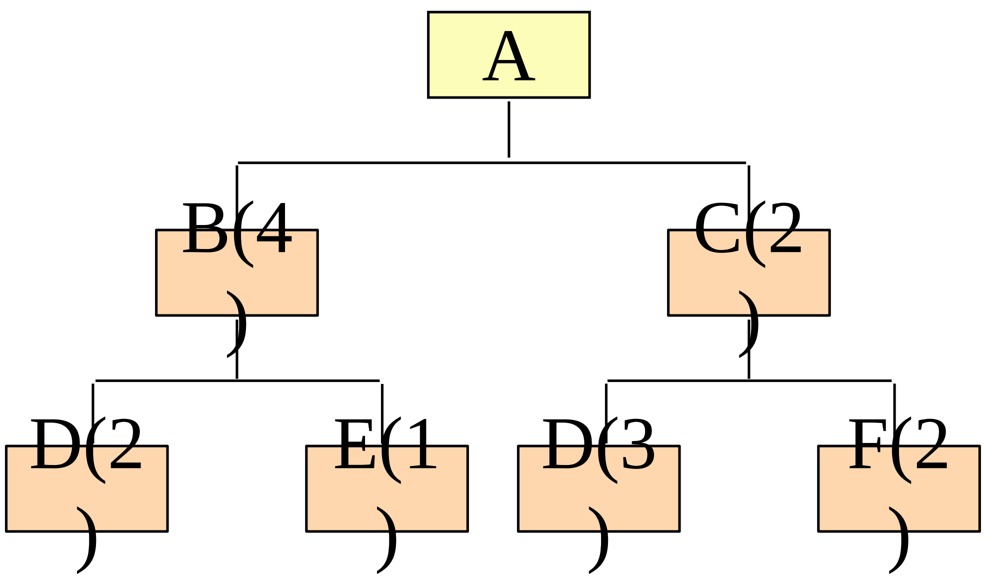
Spares

4x50=200



Finally, repeating the process for all components, we have the final materials requirements plan:

	Day:	1	2	3	4	5	6	7	8	9	10
A	Required										50
LT=1	Order Placement									50	
B	Required								20	200	
LT=2	Order Placement						20	200			
C	Required									100	
LT=1	Order Placement								100		
D	Required						55	400	300		
LT=3	Order Placement			55	400	300					
E	Required						20	200			
LT=4	Order Placement		20	200							
F	Required								200		
LT=1	Order Placement							200			



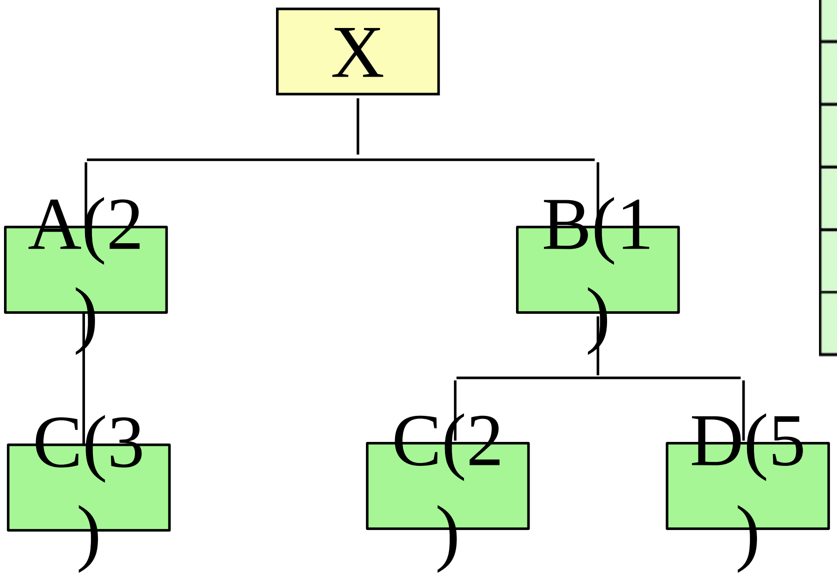
Part D: Day 6

40 + 15 spares

Additional MRP Scheduling Terminology

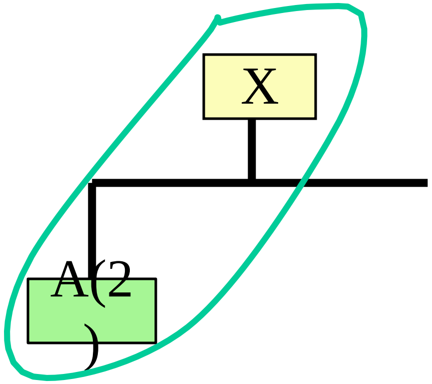
- **Gross Requirements**
- **Scheduled receipts**
- **Projected available balance**
- **Net requirements**
- **Planned order receipt**
- **Planned order release**

MRP Example



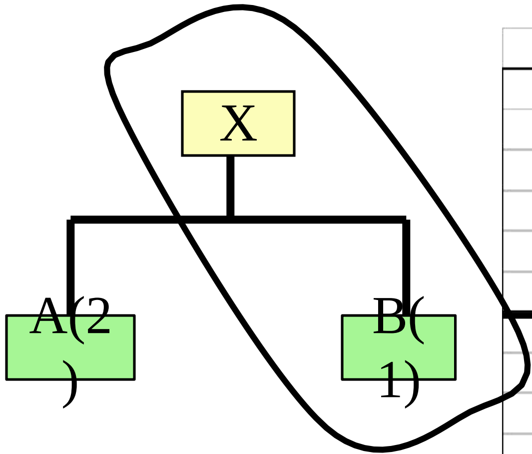
Item	On-Hand	Lead Time (Weeks)
X	50	2
A	75	3
B	25	1
C	10	2
D	20	2

Requirements include 95 units (80 firm orders and 15 forecast) of X in week 10



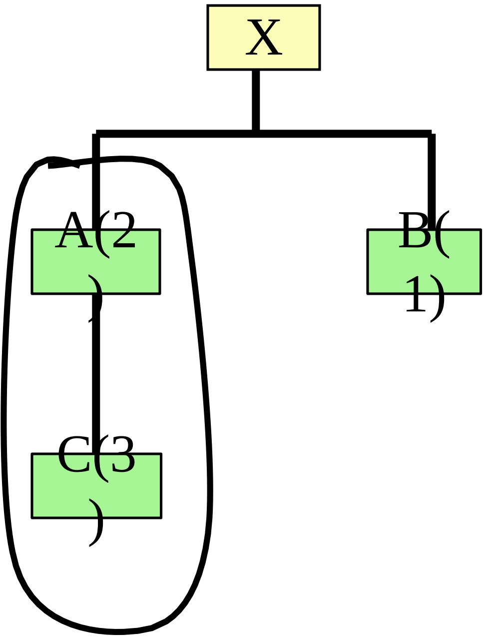
It takes 2 A's for each X

	Day:	1	2	3	4	5	6	7	8	9	10
X	Gross requirements										95
LT=2	Scheduled receipts										
	Proj. avail. balance	50	50	50	50	50	50	50	50	50	50
On-	Net requirements										45
hand	Planned order receipt										45
50	Planner order release								45		
A	Gross requirements								90		
LT=3	Scheduled receipts										
	Proj. avail. balance	75	75	75	75	75	75	75	75		
On-	Net requirements								15		
hand	Planned order receipt								15		
75	Planner order release					15					
B	Gross requirements								45		
LT=1	Scheduled receipts										
	Proj. avail. balance	25	25	25	25	25	25	25	25		
On-	Net requirements								20		
hand	Planned order receipt								20		
25	Planner order release							20			
C	Gross requirements					45		40			
LT=2	Scheduled receipts										
	Proj. avail. balance	10	10	10	10	10					
On-	Net requirements					35		40			
hand	Planned order receipt					35		40			
10	Planner order release			35		40					
D	Gross requirements							100			
LT=2	Scheduled receipts										
	Proj. avail. balance	20	20	20	20	20	20	20			
On-	Net requirements							80			
hand	Planned order receipt							80			
20	Planner order release					80					



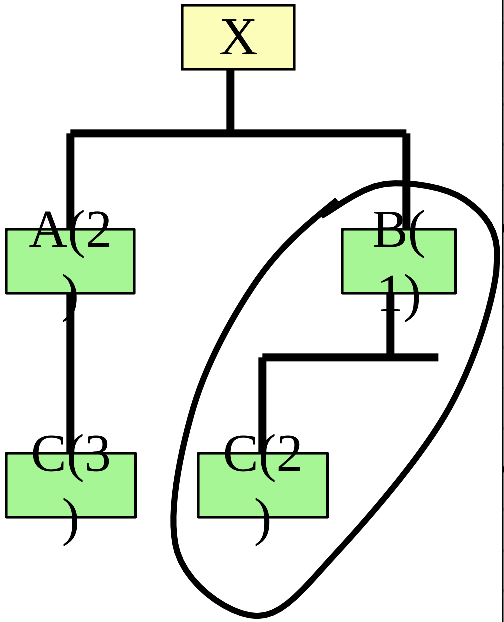
It takes 1 B for each X

	Day:	1	2	3	4	5	6	7	8	9	10
X	Gross requirements										95
LT=2	Scheduled receipts										
	Proj. avail. balance	50	50	50	50	50	50	50	50	50	50
On-	Net requirements										45
hand	Planned order receipt										45
50	Planner order release								45		
A	Gross requirements								90		
LT=3	Scheduled receipts										
	Proj. avail. balance	75	75	75	75	75	75	75	75		
On-	Net requirements								15		
hand	Planned order receipt								15		
75	Planner order release					15					
B	Gross requirements								45		
LT=1	Scheduled receipts										
	Proj. avail. balance	25	25	25	25	25	25	25	25		
On-	Net requirements								20		
hand	Planned order receipt								20		
25	Planner order release							20			
C	Gross requirements					45		40			
LT=2	Scheduled receipts										
	Proj. avail. balance	10	10	10	10	10					
On-	Net requirements					35		40			
hand	Planned order receipt					35		40			
10	Planner order release			35		40					
D	Gross requirements							100			
LT=2	Scheduled receipts										
	Proj. avail. balance	20	20	20	20	20	20	20			
On-	Net requirements							80			
hand	Planned order receipt							80			
20	Planner order release					80					



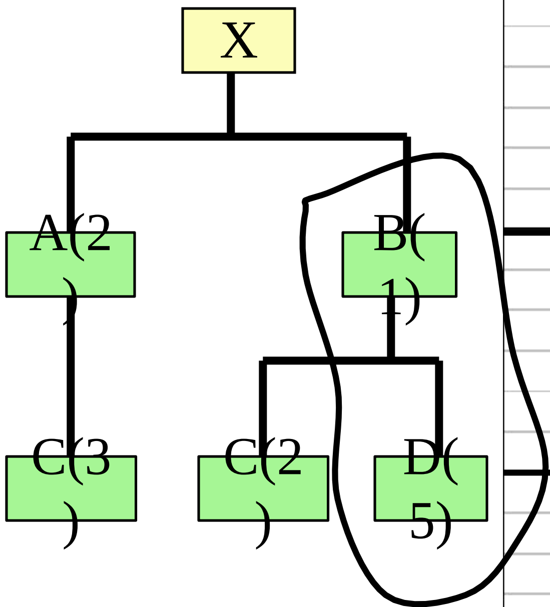
It takes 3 C’s for each A

	Day:	1	2	3	4	5	6	7	8	9	10
X	Gross requirements										95
LT=2	Scheduled receipts										
	Proj. avail. balance	50	50	50	50	50	50	50	50	50	50
On-	Net requirements										45
hand	Planned order receipt										45
50	Planner order release								45		
A	Gross requirements								90		
LT=3	Scheduled receipts										
	Proj. avail. balance	75	75	75	75	75	75	75	75		
On-	Net requirements								15		
hand	Planned order receipt								15		
75	Planner order release					15					
B	Gross requirements								45		
LT=1	Scheduled receipts										
	Proj. avail. balance	25	25	25	25	25	25	25	25		
On-	Net requirements								20		
hand	Planned order receipt								20		
25	Planner order release							20			
C	Gross requirements					45		40			
LT=2	Scheduled receipts										
	Proj. avail. balance	10	10	10	10	10					
On-	Net requirements					35		40			
hand	Planned order receipt					35		40			
10	Planner order release			35		40					
D	Gross requirements							100			
LT=2	Scheduled receipts										
	Proj. avail. balance	20	20	20	20	20	20	20			
On-	Net requirements							80			
hand	Planned order receipt							80			
20	Planner order release					80					



It takes 2 C's for each B

	Day:	1	2	3	4	5	6	7	8	9	10
X	Gross requirements										95
LT=2	Scheduled receipts										
	Proj. avail. balance	50	50	50	50	50	50	50	50	50	50
On-	Net requirements										45
hand	Planned order receipt										45
50	Planner order release								45		
A	Gross requirements								90		
LT=3	Scheduled receipts										
	Proj. avail. balance	75	75	75	75	75	75	75	75		
On-	Net requirements								15		
hand	Planned order receipt								15		
75	Planner order release					15					
B	Gross requirements								45		
LT=1	Scheduled receipts										
	Proj. avail. balance	25	25	25	25	25	25	25	25		
On-	Net requirements								20		
hand	Planned order receipt								20		
25	Planner order release							20			
C	Gross requirements					45		40			
LT=2	Scheduled receipts										
	Proj. avail. balance	10	10	10	10	10					
On-	Net requirements					35		40			
hand	Planned order receipt					35		40			
10	Planner order release			35		40					
D	Gross requirements							100			
LT=2	Scheduled receipts										
	Proj. avail. balance	20	20	20	20	20	20	20			
On-	Net requirements							80			
hand	Planned order receipt							80			
20	Planner order release					80					

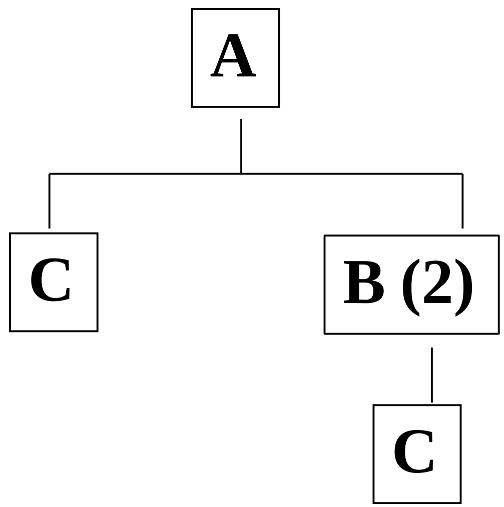


It takes 5 D’s for each B

	Day:	1	2	3	4	5	6	7	8	9	10
X	Gross requirements										95
LT=2	Scheduled receipts										
	Proj. avail. balance	50	50	50	50	50	50	50	50	50	50
On-	Net requirements										45
hand	Planned order receipt										45
50	Planner order release								45		
A	Gross requirements								90		
LT=3	Scheduled receipts										
	Proj. avail. balance	75	75	75	75	75	75	75	75		
On-	Net requirements								15		
hand	Planned order receipt								15		
75	Planner order release					15					
B	Gross requirements								45		
LT=1	Scheduled receipts										
	Proj. avail. balance	25	25	25	25	25	25	25	25		
On-	Net requirements								20		
hand	Planned order receipt								20		
25	Planner order release							20			
C	Gross requirements					45		40			
LT=2	Scheduled receipts										
	Proj. avail. balance	10	10	10	10	10					
On-	Net requirements					35		40			
hand	Planned order receipt					35		40			
10	Planner order release			35		40					
D	Gross requirements							100			
LT=2	Scheduled receipts										
	Proj. avail. balance	20	20	20	20	20	20	20			
On-	Net requirements							80			
hand	Planned order receipt							80			
20	Planner order release					80					

Numerical # 1:

Given the following product structure tree diagram, complete the MRP records for A, B, and C.



Week	4	5	6	7	8	9
Gross	**5**	**15**	**18**	**8**	**12**	**22**
Scheduled Receipt						
PAB **(21)**	21	16	1	3	15	3
P.O.Receipt			20	20		20
P.O. Release		20	20		20	
Gross		**40**	**40**		**40**	
Scheduled Receipt	**32**	32				
PAB **(20)**	20	20	12	12	12	
P.O.Receipt			40		40	
P.O. Release	40		40			
Gross	**40**	**20**	**60**		**20**	
Scheduled Receipt						
PAB **(50)**	50	10	10	10	10	10
P.O.Receipt		20	60		20	
P.O. Release	20	60		20		

PART A

Q=20; LT=1; SS=0

PART B

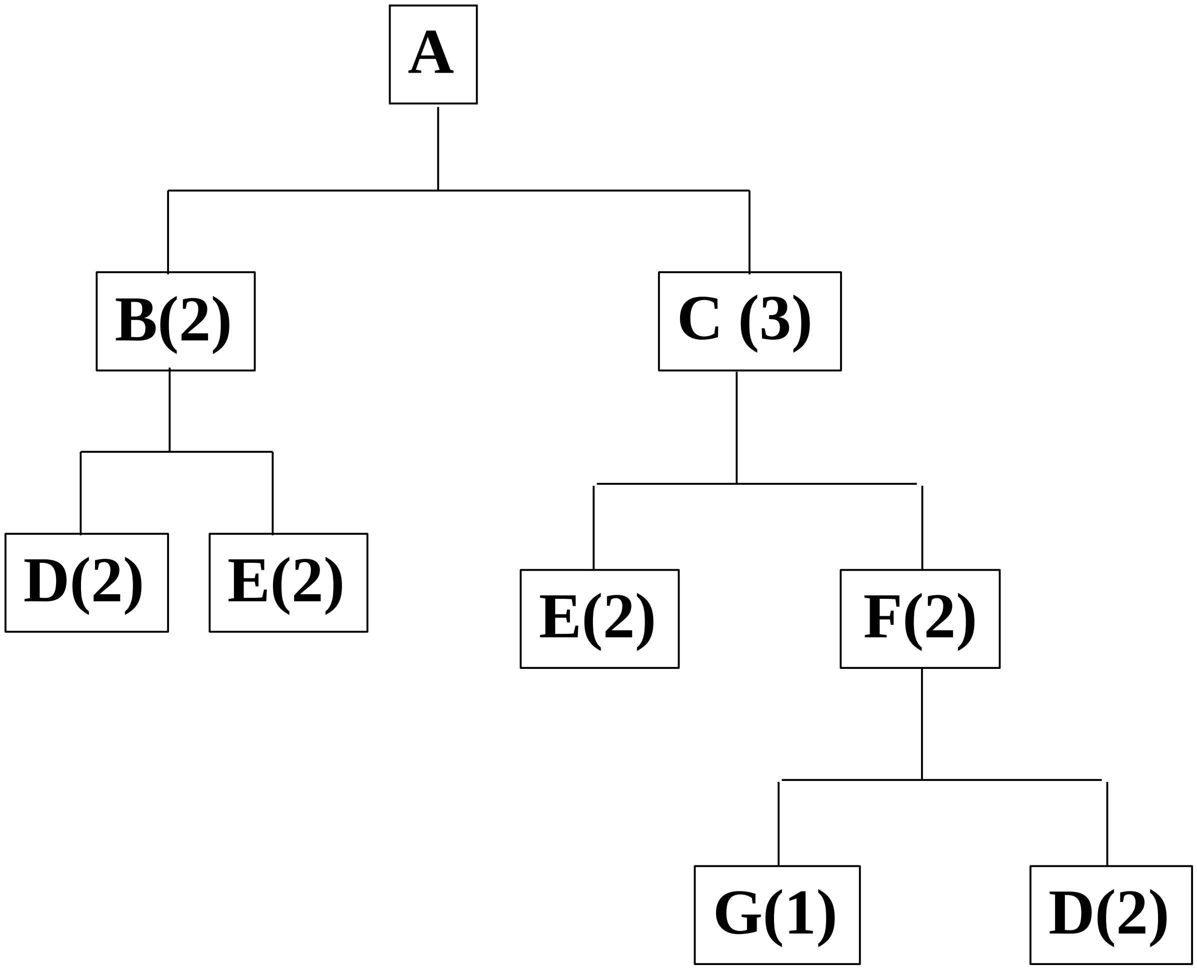
Q=40; LT=2; SS=0

PART C

Q=L-4-L; LT=1; SS=10

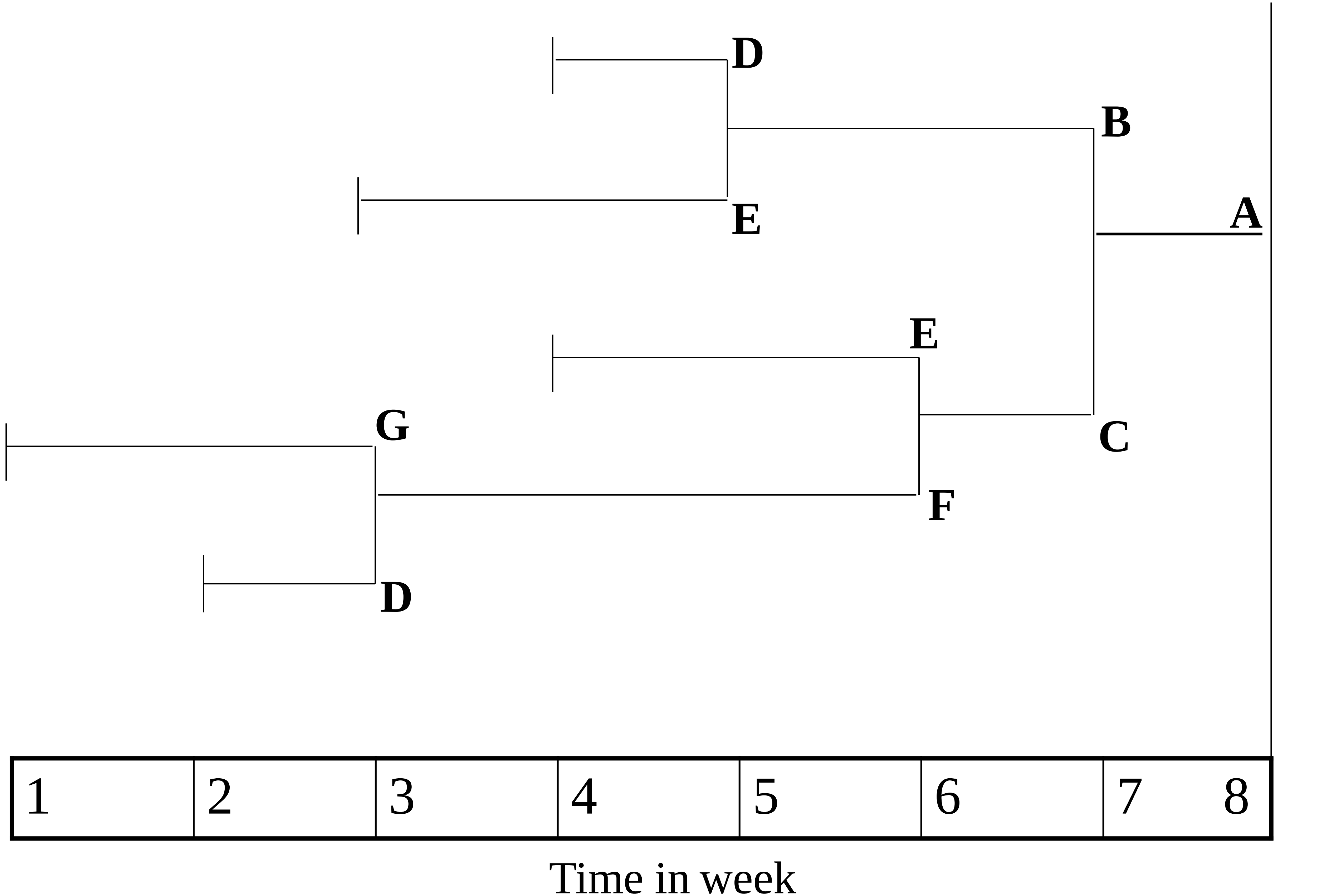
Numerical # 2:

Draw the time-phased product structure tree.



Parts	L.T.
A	1
B	2
C	1
D	1
E	2
F	3
G	2

Time-phased product structure of A



End of Session